

Date: 17/05/2017, Wednesday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q.1. Attempt any TWO of the following

- A i. Define Absorption, drug disposition, drug distribution and metabolism. [4]
ii. Write a short note on BCS classification. [6]
- B How blood flow to the kidney, Plasma concentration of drug and biological factors affect renal excretion of drug? [10]
- C Enlist the various factors influencing drug absorption and discuss the effect of drug solubility and particle size on drug absorption. [10]

Q.2. Attempt any TWO of the following

- A i. Discuss about first pass metabolism. [5]
ii. Explain *In-Vitro In-Vivo* Correlation. [5]
- B Enlist the different mechanism of drug absorption and describe any two in detail. [10]
- C What are pharmacokinetic models? Explain any one compartment model in detail. [10]

SECTION- II

Q.3. Attempt any TWO of the following

- A Define pharmacokinetics. Explain a typical plasma level time curve after single oral dose and define each term. [10]
- B Explain the different causes of non-linearity in pharmacokinetics. [10]
- C i. Define clearance, total body clearance and organ clearance. What is extraction ratio? [05]
ii. What is sink condition? Explain the significance of sink condition. [05]

Q.4. Attempt any TWO of the following

- A Explain various methods used for enhancement of bioavailability. [10]
 - B i. Explain and justify the importance of apparent volume of distribution. [08]
ii. Define bioavailability and bioequivalence. [02]
 - C Enlist methods for bioavailability measurement and explain Pharmacodynamic method. [10]
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